

triominoes. However, when we experiment, we cannot find a tiling of one of these boards using straight triominoes. A proof by exhaustion does not appear promising. Can we adapt our proof from Example 20 to prove that no such tiling exists?

Solution:

W	Blu	Blk	W	Blu	Blk	W	Blu
Blk	W	Blu	Blk	W	Blu	Blk	W
Blu	Blk	W	Blu	Blk	W	Blu	Blk
W	Blu	Blk	W	Blu	Blk	W	Blu
Blk	W	Blu	Blk	W	Blu	Blk	W
Blu	Blk	W	Blu	Blk	W	Blu	Blk
W	Blu	Blk	W	Blu	Blk	W	Blu
Blk	W	Blu	Blk	W	Blu	Blk	W

Figure 7: Coloring the squares of the standard checkerboard with three colors.

We will color the squares of the checkerboard in an attempt to adapt the proof by contradiction we gave in Example 20 of the impossibility of using dominoes to tile a standard checkerboard with opposite corners removed. Because we are using straight triominoes rather than dominoes, we color the squares using three colors rather than two colors, as shown in Figure 7. Note that there are 21 blue squares, 21 black squares and 22 white squares in this coloring.

• Next, we make the crucial observation that when a straight triomino covers three squares of the checkerboard, it covers one blue square, one black square and one white square.

• Next, note that each of the three colors appears in a corner square.

Thus, without loss of generality, we may assume that we have rotated the coloring so that the missing square is